

- 1) A charged particle of mass 5kg and 3C starts at the origin with Velocity $7a_y$ m/s and travel in a region of uniform magnetic field $B = 15a_z$ Wb/m² at $t = 5$ sec. Apply magnetostatic principle and Compute Velocity, acceleration of the particle and magnetic force on it!

sol:

Given data:

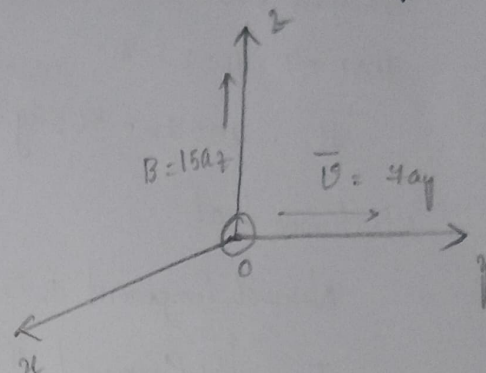
$$m = 5\text{kg} \quad q = 3\text{C} \quad \text{initial Velocity} = 7a_y \text{ m/s}$$

$$B = 15a_z \text{ Wb/m}^2 \quad t = 5\text{sec}$$

• Velocity along y direction $\rightarrow a_y$

• Magnetic field along $\rightarrow a_z$

$$a_y \times a_z = a_x \rightarrow (1)$$



We know that Force

$$\vec{F} = q(\vec{v} \times \vec{B}) = 3(7a_y) \times 15a_z = 3 \times 7 \times 15(a_y \times a_z)$$

$$= 21 \times 15(a_x)$$

$$\boxed{\vec{F} = 315a_x \text{ N}}$$

acceleration we know that

$$\vec{F} = ma$$

$$\therefore F = 315a_x$$

$$a = \frac{\vec{F}}{m} = \frac{315}{5}a_x$$

$$\boxed{a = 63a_x \text{ m/s}^2}$$

We know that Velocity equation

$$\vec{v}(t) = \vec{u} + \vec{a}t$$

$\therefore \vec{u}$ = initial Velocity

$$= 7a_y + 63a_x(5)$$

$$\boxed{= 315a_x + 7a_y \text{ m/s}}$$

2) Two homogeneous linear and isotropic media have an interface $x=0$. $x < 0$ describe medium 1 and $x > 0$ describe medium 2, $\mu_{r1} = 4$ and $\mu_{r2} = 8$ the magnetic field in medium 1 is $255a_x - 365a_y + 575a_z$ A/m. then solve for the magnetic field in medium 2, magnetic flux density in medium 1, magnetic flux density medium 2!

sol:

Given data

interface $x=0$ $x < 0 \rightarrow m1$ $x > 0 \rightarrow m2$

$$\mu_{r1} = 4, \mu_{r2} = 8$$

$$\vec{H}_1 = 255a_x - 365a_y + 575a_z \text{ A/m}$$

We know that

Normal Component B is Continuous $\boxed{B_{1n} = B_{2n}}$

Tangential Component H is Continuous $\boxed{H_{1t} = H_{2t}}$

here boundary plane ($x=0$) — Normal direction (x), Tangential (y, z)

$$\text{Normal Component} = 255a_x$$

$$\text{Tangential Component } H_{1t} = -365a_y + 575a_z$$

• Magnetic field in medium 2:

Tangential Component is same $\vec{H}_{1t} = -365a_y + 575a_z$

Normal Component $B_{1n} = B_{2n}$

$$\mu_r H_{1n} = \mu_r H_{2n}$$

$$\text{We know } B = \mu_0 H$$

$$4(255) = 8(H_{2n}) \Rightarrow H_{2n} = 127.5a_x$$

$$\boxed{\vec{H}_2 = 127.5a_x - 365a_y + 575a_z}$$

• Magnetic flux density for medium 1 & 2

$$\text{We know } B = \mu_0 H \rightarrow \textcircled{1}$$

$$\vec{B}_1 = 4\mu_0(255a_x - 365a_y + 575a_z)$$

$$\vec{B}_1 = (1020\mu_0)a_x - (1460\mu_0)a_y + (2300\mu_0)a_z \text{ Wb/m}^2$$

$$\vec{B}_2 = \mu_2 \vec{H}_2$$

$$[\because \mu_2 = \mu_{r2} \mu_0]$$

$$= \mu_{r2} \mu_0 \vec{H}_2$$

$$= 8\mu_0 (127.5\hat{a}_x - 365\hat{a}_y + 575\hat{a}_z)$$

$$\vec{B}_2 = (1020\mu_0)\hat{a}_x - (2920\mu_0)\hat{a}_y + (4600\mu_0)\hat{a}_z \text{ Wb/m}^2$$

- 3) In a magnetic flux density of $\vec{B} = 5.0\hat{a}_x + 8.0\hat{a}_y \text{ Wb/m}^2$, a current element, $14\hat{a}_z \text{ mA-m}$ is placed, then calculate the force of the current element!

Sol:

Given data

$$\vec{B} = 5.0\hat{a}_x + 8.0\hat{a}_y \text{ Wb/m}^2$$

$$\text{Current element } I d\vec{l} = 14 \times 10^{-3} \hat{a}_z \text{ A-m}$$

We know that force of current element

$$\vec{F} = I d\vec{l} \times \vec{B}$$

$$= 14 \times 10^{-3} \hat{a}_z \times (5.0\hat{a}_x + 8.0\hat{a}_y)$$

$$= 14 \times 10^{-3} [5.0(\hat{a}_z \times \hat{a}_x) + 8.0(\hat{a}_z \times \hat{a}_y)]$$

$$= 14 \times 10^{-3} [5\hat{a}_y - 8(\hat{a}_x)]$$

$$= 85 \times 10^{-3} \hat{a}_y - 136 \times 10^{-3} \hat{a}_x$$

$$= [-136\hat{a}_x + 85\hat{a}_y] 10^{-3} \text{ N}$$